Journal of Food Engineering 141 (2014) 65–73

Contents lists available at ScienceDirect

Journal of Food Engineering

journal homepage: www.elsevier.com/locate/jfoodeng

Estimation of thermo-physical properties of products with cylindrical shape during drying: The coupling between mass and heat

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A R T I C L E   I N F O

Article history:
Received 21 March 2014
Received in revised form 5 May 2014
Accepted 11 May 2014
Available online 20 May 2014

Keywords:
Heat and mass transfer
Liquid diffusion model
Shrinkage
Variable properties
Finite volume method
Evaluation of simplifications

A B S T R A C T

This article aimed at studying drying of products with cylindrical shape, using an apparent liquid diffusion model, with coupling between mass and heat at surface of the product. Model includes variable thermo-physical properties and shrinkage. The one-dimensional diffusion equation in cylindrical coordinates, with boundary condition of the third kind (i.e., convective boundary condition), was discretized by means of the finite volume method with a fully implicit formulation. The proposed solution can be used to determine thermo-physical parameters, via optimization technique, and also to simulate water migration and heat transport. Developed software was applied to describe convective drying of whole bananas. Three experiments were carried out at average drying air temperature of 47.9, 58.6 and 66.9 °C, enabling to determine the thermo-physical properties as well as to simulate the process of heat and mass transfer. The statistical indicators of the simulations made it possible to conclude that, for the three experiments, the proposed model well describes the involved processes.

1. Introduction

Food processing usually involves a stage in which heat is transferred to, and/or removed from, the product (Da Silva et al., 2011). Preservation methods such as pasteurization and cooling, among others, involve heat transfer. Thus, to describe these and other processes, the thermal properties of the product should be known. If the range of temperature involved in the process is not very large, the thermal properties can be considered constant. Thus, this type of processing is generally easy to be described because, in many occasions, the mass of the product can be also considered constant along the time as, for instance, in cooling processes. In other types of processing, such as soaking, baking and drying, simultaneous heat and mass transfer between the product and the external medium are involved. In these cases, in order to describe such processes, the thermo-physical properties of the product must be also known. This type of processing is difficult to be described, particularly the heat transfer, due to the fact that the ratio water/dry matter in the product is continually changing along the time. In the specific case of drying, water is lost along the time, while the dry matter does not vary during this process. As a consequence, in this case, several properties involved in the description of the phenomenon are variable, such as the density of the product, specific heat, thermal conductivity and thermal diffusivity (Lima et al., 2002; Mariani et al., 2008; Lemus-Mondaca et al., 2013; Perussello et al., 2013). Particularly in the drying study of fruits, the volume and moisture diffusivity of the product are also variable. As the drying operation is frequently used to prolong the shelf life of agricultural products, these properties must be known.

The post-harvest losses of agricultural products can be significantly reduced by using proper drying techniques. As above mentioned, the use of such techniques requires the knowledge of the thermo-physical properties of the agricultural product. Different approaches are used to determine thermo-physical properties and to describe water loss (and heat transfer) during drying, and some studies are presented in the following. In several drying studies, in order to describe heat and mass transfer using analytical solutions of the diffusion equation, some authors such as Ramsaroop and Persad (2012), which studied heat transfer during drying of coconut, assume the thermo-physical properties of the product with constant values. Wu et al. (2004) used numerical solutions of the diffusion equation to describe heat and mass transfer during drying of rice, but some thermo-physical properties of the product were considered constant by these researchers. However, most researchers consider these properties with variable values, and use numerical solutions of the diffusion equation to describe both: mass and heat transfer (Lima et al., 2002; Mariani et al., 2008; Lemus-Mondaca et al., 2013; Perussello et al., 2013). Usually the mass migration is described under isothermal

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http://dx.doi.org/10.1016/j.jfoodeng.2014.05.010
0260-8774/

Nomenclature

A_w, A_p, A_e, B	coefficients of the discretized diffusion equation
a, b	fitting parameters
A_s	area of the surface of the cylinder (m^2)
c_p	specific heat of product ($J\ kg^{-1}\ K^{-1}$)
c_v	specific heat of vapor of water ($J\ kg^{-1}\ K^{-1}$)
D	effective moisture diffusivity ($m^2\ s^{-1}$)
h_m	convective mass transfer coefficient ($m\ s^{-1}$)
h_T	heat transfer coefficient ($m\ s^{-1}$)
h_H	convective heat transfer coefficient h_H ($W\ m^{-2}\ K^{-1}$)
h_{fg}	latent heat of vaporization of water ($J\ kg^{-1}$)
k	thermal conductivity ($W\ m^{-1}\ K^{-1}$)
L	height of cylinder (m)
M	local moisture content ($kg_{water}\ kg^{-1}_{dry\ matter}$)
$\overline{M}$	average moisture content ($kg_{water}\ kg^{-1}_{dry\ matter}$)
m	mass of the product
N	number of control volumes
N_p	number of experimental points
r	position inside the cylinder (m)
R	radius of the cylinder (m)
R^2	coefficient of determination
t	drying time (s)
T	temperature (K)
Δr	thickness of the control volumes (m)

Subscripts

0	initial
eq	equilibrium
f	final
m, T	moisture, thermal
w, p, e	west, nodal point, east

Greek symbols

α	thermal diffusivity ($m^2\ s^{-1}$)
χ^2	chi-square
Γ^{Φ}	transport coefficient (dimension depends on the process under study)
Φ	dependent variable of the diffusion equation (dimension depends on the phenomenon under study)
λ	transport coefficient (dimension depends on the phenomenon under study)
ρ	density ($kg\ m^{-3}$)
σ_i	standard deviation of the experimental moisture content at the point i

conditions, i.e., this phenomenon is described, for each drying air temperature, in an independent way of the distribution of temperature within the product. In addition, the water migration is considered to occur in the liquid state, and its vaporization occurs at the surface where, therefore, happens the coupling between mass and heat.

Some particular considerations are made in each research on the heat and mass transfer. As examples, some interesting works are mentioned in the following. Lima et al. (2002), describing heat and mass transfer during drying of bananas, considered the density multiplied by the specific heat, in the transient term of the diffusion equation, as a variable quantity. Mariani et al. (2008), describing the heat transfer during drying of bananas, considered this quantity with a constant value along time. On the other hand, models of Lima et al. (2002) and Mariani et al. (2008) include the energy due to heating of the vapor generated at temperature of the surface up to drying air temperature. Although some other works also include this energy, most researchers neglect it. A very common simplification is presented in the following. In the calculations, most researchers use the value of the latent heat of vaporization of free water instead the value this property in the product. Wu et al. (2004) further simplifies this problem by ignoring the latent heat of vaporization of water to describe heat transfer during rice drying. In a literature search, the authors of the present paper did not find a study evaluating the implications of these simplifications above mentioned. In this sense, the aim of this article is defined in the following.

Basically, this article had two distinct objectives. First, using experimental data sets and the inverse method, a liquid diffusion model was proposed to describe heat and mass transfer during drying of whole bananas, determining their thermo-physical properties. For this purpose, the diffusion equation was numerically solved, by means of the finite volume method, with a fully implicit formulation. This solution, as well as the determination of the thermo-physical properties, permits to simulate the migration of both: mass and heat. The second objective of this article was to evaluate some simplifications proposed by researchers in the description of the heat transfer during drying.

2. Materials and methods

2.1. Drying experiments

Mature bananas *Musa acuminata*, subgroup Cavendish cv nanica was acquired from the local market, Campina Grande, Brazil. The bananas were peeled and selected by their appearance, with no evidence of mechanical damage. The experiments were carried out in a convective dryer with vertical flux, controller of temperature and controller of air velocity (Fig. 1). One sieve (number 1) with one whole banana was placed in the convective dryer, with hot air at the following average temperatures: 47.9, 58.6 and 66.9 °C. In each experiment, the drying air temperature was measured twenty times along the process, and the correspondent standard deviations for the average values of 47.9, 58.6 and 66.9 °C were 0.7, 0.8 and 0.6 °C, respectively. During the experiments, moisture content was measured by the gravimetric method. The bananas were weighed at time intervals ranging from 5 min at the beginning of drying to about 4 h at the final part of the process. At the end of each drying, the banana was removed from the dryer and placed within an oven at 105 °C. The product remained there

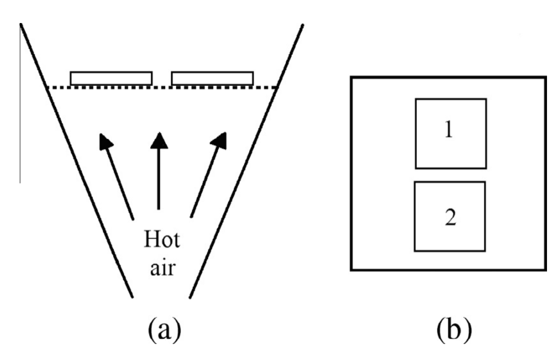

Fig. 1. Convective dryer: (a) Cross-section; (b) positions of the trays on the cross-section: number 1 – banana for mass measurement and number 2 – banana with thermocouple for temperature measurement.

Table 1
Temperature, dimension, moisture content, dry matter and specific heat of vapor for each drying air temperature.

T_{air} (°C)	L_0 (mm)	R_0 (mm)	M_0 (db)	M_f (db)	M_{eq} (db)	m_d (kg)	c_v (J kg^{-1} °C^{-1})
47.9	141	14.9	3.43	0.202	0.121	0.0192	1908
58.6	143	14.8	3.14	0.224	0.108	0.0223	1911
66.9	165	15.3	2.90	0.228	0.094	0.0276	1914

for 24 h, enabling the measurement of the dry matter. The average drying air temperatures, dimensions, moisture contents, dry matters and specific heats of the vapor are given in Table 1.

Information about the drying air (average relative humidity, $\overline{RH}$, and velocity, $\bar{v}$) and the room air (average temperature and relative humidity) is given in Table 2 for each drying air temperature. The relative humidity, velocity of the air drying as well as room temperature was measured twenty times along of each experiment, and the values provided in Table 2 are the average values obtained in each measurement process.

The radius R and height L was measured every instant in which the bananas were removed from the dryer to measure the mass. This way, the ratio between the current and initial radius, R/R_0, was determined at every instant. Expressions R/R_0 as a function of the average moisture content $\overline{M}$ were obtained using curve fitting for all drying air temperatures, as is shown in Table 3.

If the expressions R/R_0 (Table 3) are drawn as a function of the average moisture content for all drying air temperatures in a same system of axes, the obtained lines are very close one of each other. On the other hand, the measurements of mass, radius and height enabled to determine the density of the bananas along the drying process. Thus, it is possible to determine an expression for the density of the bananas as a function of the moisture content of the product.

A thermocouple was placed in the center of a second banana (sieve number 2), which was also placed in the dryer. The initial temperature of the banana is the room temperature, which is provided in Table 2. For each drying air temperature T, care was exercised to select two bananas with the same dimensions (or very close) to minimize size effect on the experimental data. The datasets of temperature were collected in regular time interval by using a digital Multimeter Cole Parmer, with temperature range of −180° to 370 °C; and accuracy of 0.1 °C in the temperature range of interest.

2.2. One-dimensional diffusion equation and its discretization

In order to describe the processes involved during drying, the following assumptions were considered: (1) the product was considered homogeneous and isotropic; (2) the water migration can be described by liquid diffusion model; (3) the mass migration is considered under isothermal conditions, since at initial instants of drying, the thermal diffusivity (10^{-7} m^2 s^{-1}) is greater than the moisture diffusivity (10^{-10} m^2 s^{-1}) and, consequently, the Biot number for heat transfer is lesser than the mass transfer Biot number; (4) the mass and thermal diffusivities are variable properties; (5) the coupling between mass and heat occurs at surface, where the liquid phase is transformed to vapor. In addition to these assumptions, it was established that radial shrinkage must be included to model.

Table 2
Relative humidity, velocity and temperature.

Drying air			Room air	
T (°C)	$\overline{RH}$ (%)	$\bar{v}$ (m/s)	T_0 (°C)	$\overline{RH}_{room}$ (%)
47.9	20.3	1.84	25.3	59.3
58.6	10.4	1.70	25.6	48.3
66.9	6.6	1.50	24.6	48.4

Table 3
Ratio R/R_0 as a function of the average moisture content.

T (°C)	R/R_0
47.9	$(0.2500\overline{M}+0.1576)^{1/3}$
58.6	$(0.2821\overline{M}+0.1123)^{1/3}$
66.9	$(0.3016\overline{M}+0.1160)^{1/3}$

For a cylinder of radius R and height L, in which $L \gg R$, the one-dimensional diffusion equation can be written in the following way:

$$\frac{\partial(\lambda\Phi)}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(r\Gamma^{\Phi}\frac{\partial\Phi}{\partial r}\right) \tag{1}$$

where Φ is the variable that describes a diffusive process (its unit depends on the specific process); t is the time (s); r defines the radial position within the cylinder (m); λ and Γ^{Φ} are the properties of the product (their units depend on the specific process). In order to numerically solve Eq. (1) using the finite volume method with fully implicit formulation (Patankar, 1980), the domain was divided into N control volumes, generating a uniform grid in the circular section of the cylinder, as is shown in Fig. 2a.

Fig. 2b shows details of an internal control volume P and its neighbors to west (W) and east (E). Integrating Eq. (1) on space, $2\pi r_P \Delta r L$, and time, from t to $t + \Delta t$, the following result is obtained:

$$\frac{\lambda_p\Phi_P - \lambda_p^0\Phi_P^0}{\Delta t} r_p\Delta r = r_e\Gamma_e^{\Phi}\frac{\partial\Phi}{\partial r}\bigg|_e - r_w\Gamma_w^{\Phi}\frac{\partial\Phi}{\partial r}\bigg|_w \tag{2}$$

in which the subscripts p, e and w represent nodal point; interface east and interface west, respectively. The term with superscript 0 is

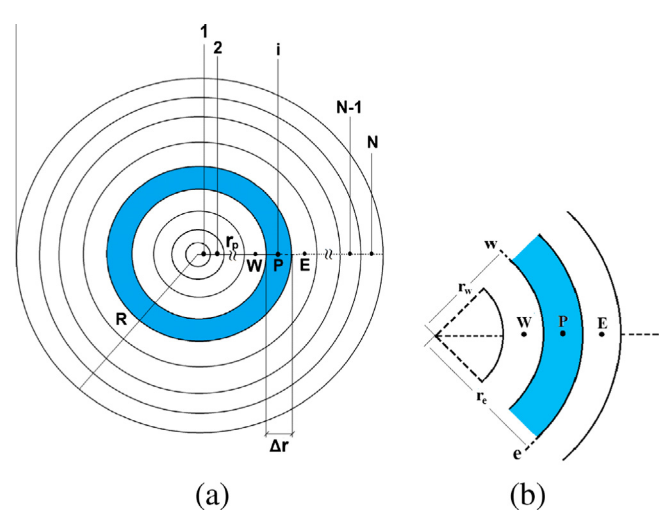

Fig. 2. (a) Uniform grid in the circular section of the infinite cylinder highlighting an internal control volume with nodal point P; (b) details of the control volume P and its neighbors to west (W) and east (E).

calculated at beginning of the time interval (t), while the terms without this superscript are calculated at the end ($t + \Delta t$). Eq. (2) is a partial result that will be used to obtain algebraic equations for each type of control volume as is shown in the following.

2.2.1. Internal control volumes

For this type of control volume, with neighbors to west and east, the derivatives of Eq. (2) can be given in the following way:

$$\left.\frac{\partial \Phi}{\partial r}\right|_e \cong \frac{\Phi_E - \Phi_P}{\Delta r} \quad \left.\frac{\partial \Phi}{\partial r}\right|_w \cong \frac{\Phi_P - \Phi_W}{\Delta r} \tag{3a–b}$$

Substituting Eq. (3a–b) into Eq. (2), the following algebraic equation is obtained:

$$A_p\Phi_P + A_w\Phi_W + A_e\Phi_E = B \tag{4}$$

in which

$$A_w = -\frac{r_w}{\Delta r}\frac{\Gamma_w^\Phi}{\lambda_P} \quad A_p = \frac{r_P\Delta r}{\Delta t} + \frac{r_e}{\Delta r}\frac{\Gamma_e^\Phi}{\lambda_P} + \frac{r_w}{\Delta r}\frac{\Gamma_w^\Phi}{\lambda_P}$$
$$A_e = -\frac{r_e}{\Delta r}\frac{\Gamma_e^\Phi}{\lambda_P} \quad B = \frac{\lambda_P^0 r_P\Delta r}{\lambda_P\Delta t}\Phi_P^0 \tag{5a–d}$$

2.2.2. Control volume number 1

For the control volume at the center of the grid, due to the symmetry, the west mass flux is zero. Consequently, the derivative given in Eq. (3b) is zero. Thus, for control volume number 1 it is obtained:

$$A_p\Phi_P + A_e\Phi_E = B \tag{6}$$

where

$$A_p = \frac{r_P\Delta r}{\Delta t} + \frac{r_e}{\Delta r}\frac{\Gamma_e^\Phi}{\lambda_P} \quad A_e = -\frac{r_e}{\Delta r}\frac{\Gamma_e^\Phi}{\lambda_P} \quad B = \frac{\lambda_P^0 r_P\Delta r}{\lambda_P\Delta t}\Phi_P^0 \tag{7a–c}$$

2.2.3. Water removal and heat transport

In order to apply the above results in the water removal, in Eq. (1) it should be imposed: $\Phi \equiv M$, in which M is the moisture content in dry basis ($\mathrm{kg_{water}\,kg^{-1}_{dry\,matter}}$); $\lambda = 1$ (dimensionless); $\Gamma^\Phi = D$, i.e., effective moisture diffusivity ($\mathrm{m^2\,s^{-1}}$).

As an observation, in order to describe heat transport, the thermal properties referring to advection, radiation, phase change, heat sources and conduction should be known. Many times, solely the conduction mechanism and phase change are used to describe heat transfer and, consequently, the properties involved (such as thermal diffusivity and conductivity) are considered as "apparent". However, in this article, for simplicity, the thermal diffusivity will continue to be named with this same name.

In the heat conduction study, the imposition on the variables of Eq. (1) Φ, λ and Γ^Φ are as follows: $\Phi \equiv T$, i.e., temperature (K); $\lambda = \rho c_p$, where ρ is the density ($\mathrm{kg\,m^{-3}}$) and c_p is the specific heat of the product ($\mathrm{J\,kg^{-1}\,K^{-1}}$); $\Gamma^\Phi = k$, i.e., thermal conductivity ($\mathrm{W\,m^{-1}\,K^{-1}}$). If the thermal conductivity is divided by ρc_p, a thermal property called thermal diffusivity ($\mathrm{m^2\,s^{-1}}$) is obtained: $\alpha = k/\rho c_p$.

For the control volume N, which is in contact with the external medium, two different equations must be obtained to complete the discretization of the diffusion equation to describe mass and heat transfer. These equations are obtained in the following.

2.2.4. Water removal: control volume N at boundary of the cylinder

For the control volume N, supposing boundary condition of the third kind, the following discretized equation is obtained:

$$A_pM_P + A_wM_W = B \tag{8}$$

where

$$A_p = \frac{r_P\Delta r}{\Delta t} + \frac{r_eD_e}{\frac{D_e}{h_m} + \frac{\Delta r}{2}} + \frac{r_w}{\Delta r}D_w \quad A_w = -\frac{r_w}{\Delta r}D_w \quad B = \frac{r_P\Delta r}{\Delta t}M_P^0 + \frac{r_eD_e}{\frac{D_e}{h_m} + \frac{\Delta r}{2}}M_{eq} \tag{9a–c}$$

in which h_m is the convective mass transfer coefficient ($\mathrm{m\,s^{-1}}$) and M_{eq} is the equilibrium moisture content (db). Detailed steps on the obtaining of Eq. (9a–c) for moisture migration are presented by Silva et al. (2012a).

2.2.5. Heat transport: control volume N at boundary of the cylinder

In this article, apparent liquid diffusion was assumed. Thus, liquid water migrates from internal medium to the surface of the product, where occurs change to vapor. Thus, for the control volume N, the boundary condition of the third kind is expressed in the following way:

$$-\alpha_e\left.\frac{\partial T}{\partial r}\right|_e = h_T(T_e - T_{eq}) - \frac{[h_{fg} + c_v(T_e - T_{eq})]\frac{dm}{dt}}{\rho c_P A_s} \tag{10}$$

where h_T is the heat transfer coefficient ($\mathrm{m\,s^{-1}}$); T_{eq} is the equilibrium temperature (K); c_v is specific heat of the vapor ($\mathrm{J\,kg^{-1}\,K^{-1}}$); h_{fg} is the latent heat of vaporization ($\mathrm{J\,kg^{-1}}$); dm/dt is the water loss rate ($\mathrm{kg\,s^{-1}}$) and A_s is the area of the surface of the cylinder ($\mathrm{m^2}$), given by $A_s = 2\pi RL$. In this article, it was assumed that the ratios L/L_0 are given by the same expressions provided in Table 3 for the radius. The convective heat transfer coefficient h_H ($\mathrm{W\,m^{-2}\,K^{-1}}$) is related with h_T through the expression $h_T = h_H/(\rho c_p)$. On the other hand, combining the result to be obtained with the discretization of Eq. (10) with Eq. (2) applied to heat transfer, Eq. (11) is determined:

$$A_pT_P + A_wT_W = B \tag{11}$$

in which

$$A_p = \frac{r_P\Delta r}{\Delta t} + \frac{\alpha_e r_e}{\frac{\alpha_e}{h_T - \frac{c_v}{\rho c_P A_s}\frac{dm}{dt}} + \frac{\Delta r}{2}} + \frac{r_w}{\Delta r}\alpha_w \quad A_w = -\frac{r_w}{\Delta r}\alpha_w$$

$$B = \frac{\rho^0 c_P^0 r_P\Delta r}{\rho c_p\Delta t}T_P^0 + \frac{\alpha_e r_e}{\frac{\alpha_e}{h_T - \frac{c_v}{\rho c_P A_s}\frac{dm}{dt}} + \frac{\Delta r}{2}}T_{eq} + \frac{\alpha_e r_e\frac{h_{fg}}{\rho c_P A_s}\frac{dm}{dt}}{\alpha_e + \frac{\Delta r}{2}\left(h_T - \frac{c_v}{\rho c_P A_s}\frac{dm}{dt}\right)} \tag{12a–c}$$

2.3. Mass and thermal diffusivity

For the nodal points, the diffusivity Γ^Φ (D or α) can be calculated from an appropriate relation between such property and the value of Φ,

$$\Gamma^\Phi = f(a, b, \Phi) \tag{13}$$

where a and b are parameters that fit the numerical solution to the experimental data, and they can be determined by optimization.

On the interfaces of the internal control volumes, for example e, for uniform grids the following expression (harmonic mean) should be used to determine Γ^Φ (Patankar, 1980):

$$\Gamma_e^\Phi = \frac{2\Gamma_E^\Phi\Gamma_P^\Phi}{\Gamma_E^\Phi + \Gamma_P^\Phi}, \tag{14}$$

If the diffusivity is constant, the coefficients A of the discretized equations are calculated only once, and B is calculated in each time step because its value depends on Φ_P^0, which is the value of Φ in the control volume P at the initial instant of each time step. On the other hand, if Γ^Φ is variable, the coefficients A are also calculated in each time step, due to the nonlinearities caused by the variation of such parameter. In this case, if the time refinement is adequate, errors due to the nonlinearities can be discarded.

In this article, the function $f(a,b,M)$ used to relate the effective mass diffusivity with the local value of the moisture content is given by (Silva et al., 2012a):

$$D = b_m \exp(a_m M), \tag{15}$$

where a_m and b_m are parameters to be determined by optimization and M is the local moisture content in dry basis.

It is well known that the thermal diffusivity of a product depends on its composition. In this sense, the thermal diffusivity of fresh fruits is very high, due to their high moisture content. During drying, the thermal diffusivity of a fruit at any instant depends on its water quantity. As water is continually lost during drying, i.e., the moisture content of the product is diminishing, the value of the thermal diffusivity also diminishes along time (Lima et al., 2002; Mariani et al., 2008). Thus, in this article, for each drying air temperature, the following expression was proposed for this property:

$$\alpha = b_T \exp(a_T \overline{M}^2), \tag{16}$$

in which a_T and b_T are constants for each drying air temperature, and are determined by optimization. On the other hand, $\overline{M}$ is the average moisture content of the fruit at any instant during the drying process.

2.4. General considerations

The water loss rate, dm/dt, necessary to describe heat transport, can be determined when water migration is described. Thus, initially water migration was studied. In each time step, a system of equations (Eqs. (4), (6), and (8)) is solved using Tridiagonal Matrix Algorithm method, TDMA (Press et al., 1996). Since water migration is described, in each time step the water loss rate is determined by:

$$\frac{dm}{dt} = \frac{m(t+\Delta t) - m(t)}{\Delta t} \tag{17}$$

in which the mass of water (kg) at instant t is given by $m(t) = m_d\overline{M}(t)$ where m_d is the mass of the dry matter (kg), given in Table 1, and $\overline{M}(t)$ is the average moisture content in dry basis ($\text{kg}_{\text{water}}\ \text{kg}^{-1}_{\text{dry matter}}$), calculated by:

$$\overline{M}(t) = \frac{\sum_{i=1}^{N} M_i r_i}{\sum_{i=1}^{N} r_i} \tag{18}$$

where M_i is the moisture content in the control volume "i" and r_i is its radius (m). Due to the shrinkage, in each time step the radius of the cylinder varies and, therefore, its value must be recalculated, as well as the thickness Δr of the control volumes of the uniform grid. Expressions for these calculations are given in Table 3.

The latent heat of water vaporization at the surface of the bananas was calculated according to the expression provided by Silva et al. (2012b):

$$h_{fg} = 2529.1 M^{(-1.782\times10^{-2} - 3.570\times10^{-4}T)} - 2.386T \tag{19}$$

where h_{fg} is obtained in kJ kg^{-1} when M is given in dry basis and T in °C.

For a given time step, in order to use Eqs. (4), (6), and (11) to determine the temperature in each control volume, the specific heat and density must be known at beginning and final of each time step. This is obvious because the coefficients B of these equations depend on these values. The specific heat of the bananas was estimated according to the expression provided by Sweat (1986) for agricultural products:

$$c_p = 1381 + 2930X \tag{20}$$

where X is the average moisture content, in wet basis (decimal), and c_p is obtained in J kg^{-1} K^{-1}.

Specific heat for the water vapor, c_v, is provided in Table 1. On the other hand, with the measurements accomplished in the experiments, it was possible to determine an expression for the density ρ of the bananas (kg m^{-3}) as a function of the average moisture content, $\overline{M}$ (db), by curve fit:

$$\rho = 1533 e^{-0.1704\overline{M}} \tag{21}$$

Thus, in each time step, if the values of the thermal diffusivity and heat transfer coefficient are provided in Eqs. (4), (6), and (11), the obtained system can be solved to determine the temperature in each control volume. At each time step, the system of equations was solved by using of TDMA method. Similar comment is also valid to study mass migration. Note that, if the thermal diffusivity and heat transfer coefficient are unknown, the parameters a_T, b_T and h_T can be determined by optimization (inverse method) using experimental data set, as is presented in the following.

2.5. Optimization

In order to determine the parameters Γ^{Φ} (D or α) and h (h_m or h_T), the parameters a and b of the function $\Gamma^{\Phi} = f(a, b, \Phi)$ and h can be determined by optimization using an experimental dataset. The expression of the chi-square (Bevington and Robinson, 1992; Taylor, 1997) was chosen as objective function:

$$\chi^2 = \sum_{j=1}^{N_p} \left[\overline{\Phi}_j^{\exp} - \overline{\Phi}_j^{sim}\right]^2 \frac{1}{\sigma_j^2} \tag{22}$$

where $\overline{\Phi}_j^{\exp}$ is the value of Φ measured in the experimental point "j"; $\overline{\Phi}_j^{sim}$ is the correspondent average value of Φ obtained by simulation; N_p is the number of experimental points and $1/\sigma_j^2$ is the statistical weight referring to the point "j". If the statistical weights are unknown, they can be made equal to a common value, for instance 1. In Eq. (22), the chi-square depends on $\overline{\Phi}_j^{sim}$, which depends on Γ^{Φ} (i.e., parameters a_m or a_T and b_m or b_T) and h (h_m or h_T). Thus, the parameters a, b and h can be determined by minimization of Eq. (22) through successive trials.

3. Results and discussion

The circular section of the bananas was divided into 100 control volumes and the drying time was divided into 5000 steps. An unreported study on grid and time refinement indicated that such values are adequate to describe two processes: mass and heat transfer.

3.1. Results for water migration

The effective moisture diffusivity was considered variable as a function of the local moisture content as is given by Eq. (15), and the radius was also considered variable, according to Table 3. Thus, the parameters related with water transfer, obtained by optimization, are presented in Table 4.

The simulations of the drying kinetics are shown in Fig. 3a–c, and the superposition of these simulations is shown in Fig. 3d.

As additional information, the graphs of Fig. 3a–c were generated by own software developed to perform the optimizations.

Table 4
Parameters of the water migration obtained by optimization.

T (°C)	a_m	b_m (m^2 min^{-1})	h_m (m min^{-1})	R^2	χ^2
47.9	0.687	15.4×10^{-9}	2.66×10^{-5}	0.99970	15.6×10^{-3}
58.6	0.785	19.4×10^{-9}	2.68×10^{-5}	0.99989	3.87×10^{-3}
66.9	0.821	28.9×10^{-9}	2.99×10^{-5}	0.99964	9.20×10^{-3}

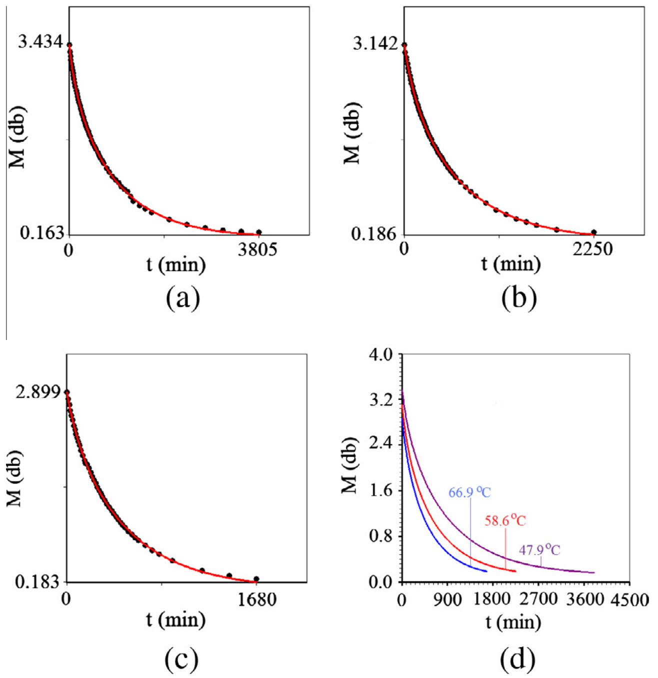

Fig. 3. Simulation of water migration at drying air temperature of: (a) 47.9 °C; (b) 58.6 °C; (c) 66.9 °C; (d) superposition of the simulated curves.

On the other hand, with the results from Table 4, the graphs of the functions $D(M)$, given by Eq. (15), are presented in Fig. 4 for three temperatures of the drying air.

In order to present the moisture distributions within the circular section that represents the bananas, Fig. 5 highlights these distributions at 200 min, for three temperatures of the drying air.

3.2. Results for heat transport

Thermal diffusivity was considered variable as a function of the average moisture content, as is given by Eq. (16). In the heat transport study, the radius was also considered variable, according to Table 3. Thus, the parameters related with the heat transfer, obtained by optimization, are presented in Table 5.

The simulations of the heating kinetics in the center of the bananas are shown in Fig. 6a–c, while Fig. 6d presents the superposition of these simulations.

Note that the graphs of Fig. 6a–c were also generated by software developed to perform the optimization processes.

With the results from Table 5, the graphs of the functions $\alpha(\overline{M})$, given by Eq. (16), are presented in Fig. 7, for three temperatures of the drying air.

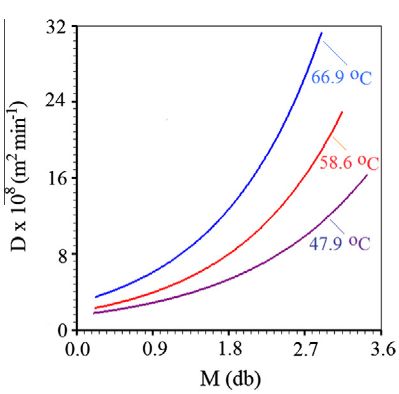

Fig. 4. Effective moisture diffusivity as a function of the local moisture content at temperature of: (a) 47.9 °C; (b) 58.6 °C; (c) 66.9 °C.

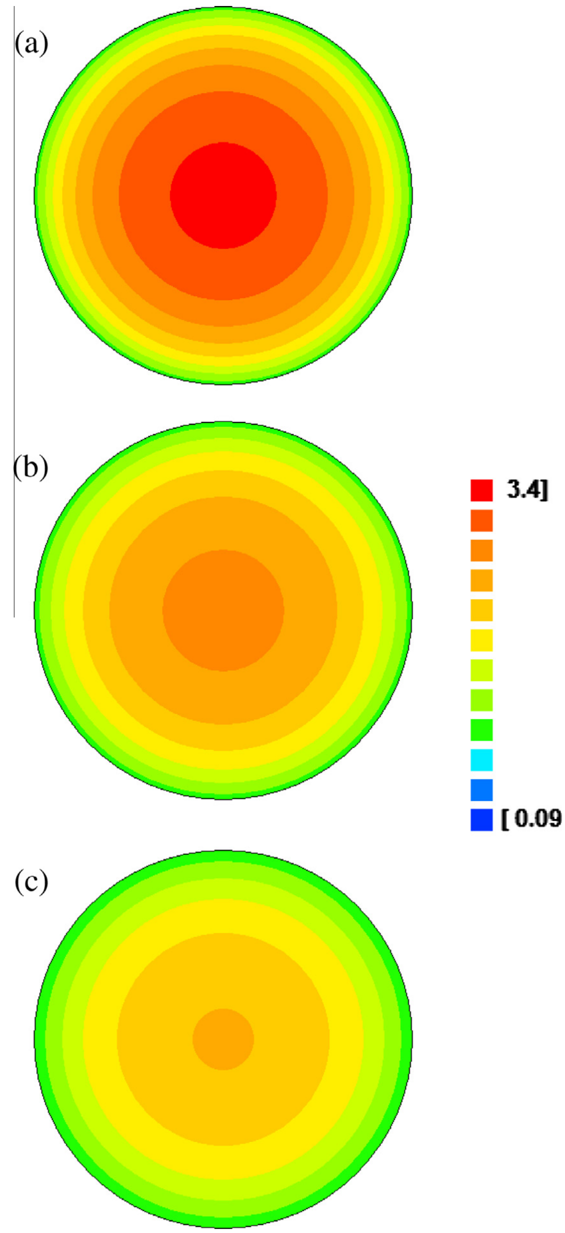

Fig. 5. Distribution of the moisture content at $t = 200$ min for the drying air temperature of: (a) 47.9 °C; (b) 58.6 °C; (c) 66.9 °C.

Table 5
Parameters of the heat conduction obtained by optimization.

T (°C)	a_T	b_T (m^2 min^{-1})	h_T (m min^{-1})	R^2	χ^2
47.9	0.313	1.26×10^{-7}	5.61×10^{-4}	0.99071	14.3
58.6	0.398	1.06×10^{-7}	5.71×10^{-4}	0.98843	28.9
66.9	0.520	1.26×10^{-7}	9.49×10^{-4}	0.98084	33.4

The distributions of temperature within the circular section that represents the bananas at $t = 10$ min are given in Fig. 8 for three temperatures of the drying air.

3.3. Analysis of simplifications for the heat transfer problem

3.3.1. Exclusion of the heating of the vapor

In the proposed model, vapor is generated at temperature of the surface and is heated to the drying air temperature. In order to test the influence of the heating of the vapor, a simulation was

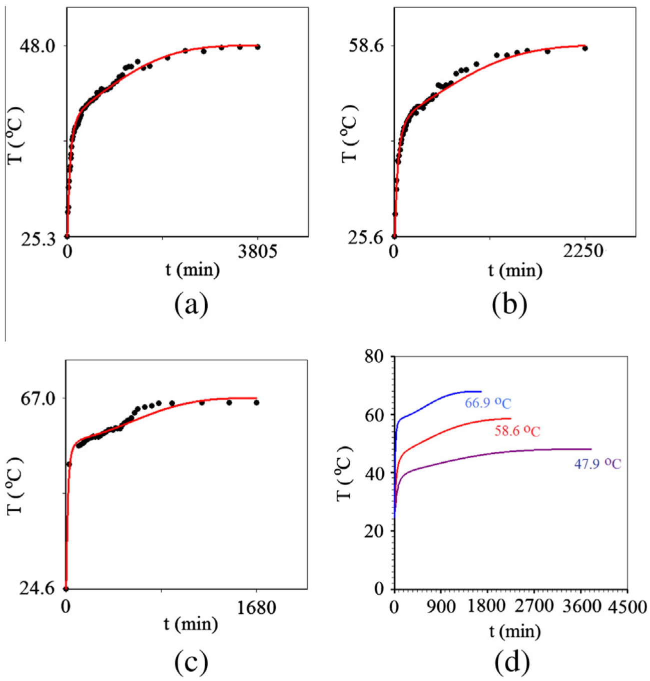

Fig. 6. Temperature in the center of the banana at drying air temperature of: (a) 47.9 °C; (b) 58.6 °C; (c) 66.9 °C; (d) superposition of the simulated curves.

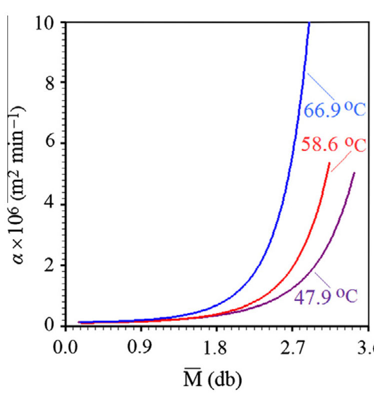

Fig. 7. Thermal diffusivity as a function of the average moisture content at drying air temperature of: (a) 47.9 °C; (b) 58.6 °C; (c) 66.9 °C.

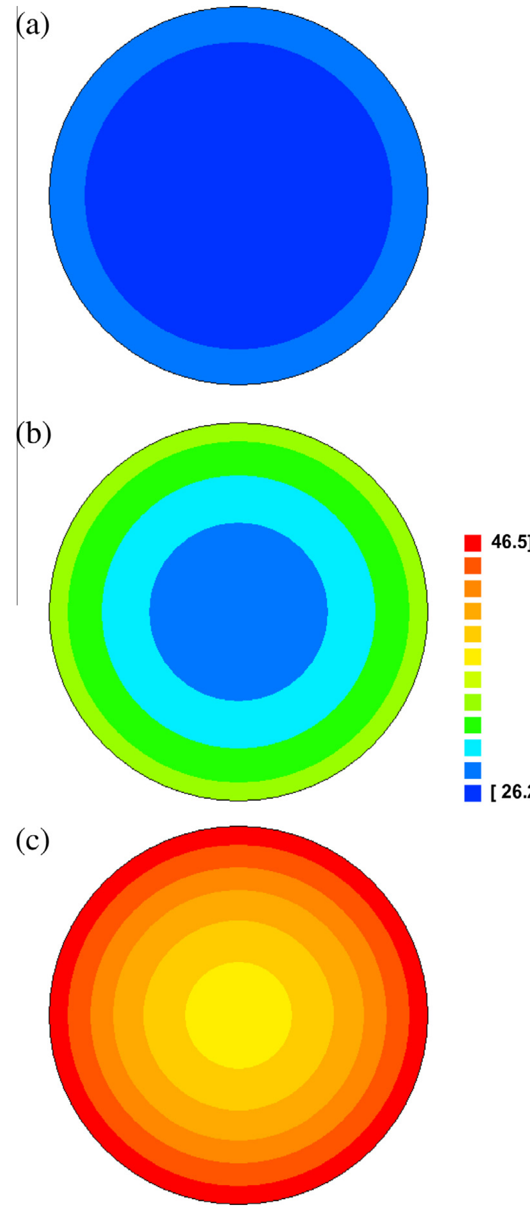

Fig. 8. Distribution of the temperatures at t = 10 min for the drying air temperature of: (a) 47.9 °C; (b) 58.6 °C; (c) 66.9 °C.

performed at 58.6 °C, considering the parameters given in Table 5, but supposing $c_v = 0$. In this case, the statistical indicators of the simulation were $\chi^2 = 29.2$ (instead 28.9) and $R^2 = 0.98835$ (instead 0.98843). A graph showing the simulation of the heating kinetics at the center of the banana does not present a significant visual difference when compared with Fig. 6b and, therefore, was not presented herein. Similar results were obtained for the drying air temperatures of 47.9 and 66.9 °C.

3.3.2. Latent heat of water vaporization

In a simulation with the drying air temperature at 58.6 °C, if the results obtained in Table 5 are used, but latent heat of water vaporization is neglected ($h_{fg} = 0$), the statistical indicators are: $\chi^2 = 4146.1$ (instead 28.9) and $R^2 = 0.75502$ (instead 0.98843). These results are completely unacceptable, and similar results are obtained for the drying air temperatures of 47.9 and 66.9 °C. On the other hand, at 58.6 °C, the use of the value of the latent heat of vaporization of free water in the simulation resulted in statistical indicators given by $\chi^2 = 29.6$ (instead 28.9) and $R^2 = 0.98832$ (instead 0.98843). Similar results were obtained at 47.9 and 66.9 °C.

3.3.3. Constant density and specific heat of bananas

In many occasions, the transient term of Eq. (1) applied to heat transfer, given by $\partial(\rho c_p T)/\partial t$, is written as $\rho c_p \partial T/\partial t$, i.e., in this term the product ρc_p is considered constant. In order to test this simplification for the drying case, the results obtained in Table 5 for $T = 58.6$ °C were used in a simulation, considering the transient term given by $\rho c_p \partial T/\partial t$. As result, the chi-square was 264.1 (instead 28.9) and the determination coefficient was 0.97835 (instead 0.98843). Similar results were also obtained for the drying air temperatures of 47.9 and 66.9 °C, indicating that this simplification should be avoided in drying studies.

3.4. Discussion

Basically, in the transient diffusion study, two methods are used to determine the thermal diffusivity and heat transfer coefficient of agricultural products: empirical correlations (Karim and Hawlader, 2005; Perussello et al., 2013) and optimization (Lima et al., 2002;

Silva et al., 2011, 2013). In the present article, only optimizations were used to determine α and h_T, and the obtained results are compatible with those reported in the literature for bananas (Lima et al., 2002; Mariani et al., 2008).

Trough an inspection of Table 4 it is possible to affirm that the statistical indicators to describe the mass transfer are excellent. Particularly, the determination coefficients are greater than 0.99960. Similar conclusion on the adequacy of the proposed model can be obtained by an observation of Fig. 3a–c for three temperatures of the drying air. On the other hand, Fig. 3d presents a superposition of the drying kinetics, in which it is possible to evaluate how much one simulation is faster than other. From Table 4 and Eq. (15), the minimum and maximum values of the effective moisture diffusivity are $2.79 \times 10^{-10}\ \mathrm{m^2\ s^{-1}}$ (47.9 °C and $M = 0.121$ db) and $5.21 \times 10^{-9}\ \mathrm{m^2\ s^{-1}}$ (66.9 °C and $M = 2.90$ db). Details on effective moisture diffusivity as a function of the local moisture content are presented in Fig. 4. In this figure it is realized that, the more the temperature of the drying air is high, the greater the effective moisture diffusivity. This obtained result is a consequence of the following fact: increasing the drying air temperature increases the overall product temperature and, hence, the moisture diffusivity. On the other hand, the convective mass transfer coefficient varied from 4.43×10^{-7} to $4.98 \times 10^{-7}\ \mathrm{m\ s^{-1}}$. Similar results were obtained by Silva et al. (2012a) and Lima et al. (2002) studying drying of bananas.

The moisture distributions are presented in Fig. 5, with no scale, for three drying air temperatures, at $t = 200$ min. An observation of this figure allows to conclude that the more the temperature of the drying air is high, the greater the loss of moisture. Although this conclusion is quite obvious, the calculations performed herein are necessary for a subsequent determining the stress within the fruit due to the difference in the moisture concentration.

Table 5 presents the results of the optimizations for the temperatures measured in the center of the bananas. The statistical indicators show that the obtained results are reasonable. Particularly, the determination coefficients are greater than 0.98000. A visual observation of Fig. 6a–c makes it possible to conclude that the simulated lines agree with experimental datasets. On the other hand, Fig. 6d enables to realize that the heating process is very fast. Differently of water migration, few minutes are enough so that the temperature at center of the fruit approximates of its equilibrium value. This fact permitted to establish the assumption number 3 of this article: "the mass migration is considered under isothermal conditions, since at initial instants of drying the thermal diffusivity ($10^{-7}\ \mathrm{m^2\ s^{-1}}$) is greater than the moisture diffusivity ($10^{-10}\ \mathrm{m^2\ s^{-1}}$)".

Inspection of Fig. 7 indicates that, for all temperatures of the drying air, the thermal diffusivity exponentially decreases with the decreasing of average moisture content of the banana. Shyamal et al. (1994), studying the thermal properties of the wheat, concluded that the thermal diffusivity decrease linearly with moisture content. However, the range of the moisture content in drying of grains is lesser than the range of fruits such as bananas. A similar result to the obtained herein for thermal diffusivity of bananas was obtained by Mariani et al. (2008). However, the expressions proposed by these researchers for the thermal diffusivity involved three and four parameters and, in the present article, only two parameters were involved. In the present article, at the beginning of drying, the values for the thermal diffusivities are given by 0.835×10^{-7} (47.9 °C), 0.894×10^{-7} (58.6 °C) and $1.67 \times 10^{-7}\ \mathrm{m^2\ s^{-1}}$ (66.9 °C). Based on the literature, these results are acceptable for agricultural products with high moisture content, such as bananas. At the end of the drying processes, the values obtained in this work for thermal diffusivity are of about $2.0 \times 10^{-9}\ \mathrm{m^2\ s^{-1}}$. The highest value obtained in the present paper for the thermal diffusivity ($1.67 \times 10^{-7}\ \mathrm{m^2\ s^{-1}}$) is lower that those obtained by Mariani et al. (2008), 1.88×10^{-7}, and Lima et al. (2002), $2.70 \times 10^{-7}\ \mathrm{m^2\ s^{-1}}$. On the other hand, the value obtained herein is closest to the value of the thermal diffusivity of the water, as is expected. In addition, the lower value obtained in this article, at the end of drying, is compatible with two works abovementioned.

In their article, Mariani et al. (2008) have determined the values for convective heat transfer coefficient based on the Nusselt number, i.e., using empirical correlations instead optimization. Their results were from 15 to $35\ \mathrm{W\ m^{-2}\ °C^{-1}}$. In the present work, this property was determined by optimization and, at beginning of the process, ranged between 15.2 (47.9 °C) and $28.3\ \mathrm{W\ m^{-2}\ °C^{-1}}$ ($66.9\ °C^{-1}$). These results are compatible with those from Mariani et al. (2008), and also agree with those obtained by Lima et al. (2002): from 12.7 to $19.0\ \mathrm{W\ m^{-2}\ °C^{-1}}$.

From Fig. 8 it is possible to realize that, at instant $t = 10$ min, the range of temperature in the interior of the banana increases with the increasing of the drying air temperature. Remembering that bananas are at room temperature at the initial instant, at drying air temperature of 47.9 °C and $t = 10$ min the range of temperature in the circular section was from 26.2 (center) to 28.9 °C (surface). For the drying air temperatures of 58.6 and 66.9 °C, these ranges were 28.4–34.0 and 37.3–46.4 °C, respectively. Finally, although the three circles of Fig. 8 have the same size, i.e., three figures are drawn with no scale, it is expected that the circle related to drying at 66.9 °C has lesser radius than other circles, once the banana at this temperature dries faster than the other two cases. Obviously, the shrinkage due to water loss is more important than the effect of expansion by temperature increase.

Lima et al. (2002) and Mariani et al. (2008), in their analyses, included the specific heat of the water vapor in order to describe heat transport. However, several researchers such as Rinaldi et al. (2011) and also Ramsaroop and Persad (2012), studying simultaneous heat and mass transfer, did not consider this phenomenon. The simulations performed in the present work made it possible to realize that the heating of vapor at surface temperature to drying air temperature is not significant in the description of the heat transport during convective drying of bananas. On the other hand, Wu et al. (2004) studied heat and mass transfer during drying of rice. These researchers used two diffusion equations to describe water and heat migration, supposing that two phenomena are independent, i.e., the latent heat of water vaporization was neglected. However, the results obtained in the present article indicated that the coupling between two phenomena is very important and the latent heat of water vaporization cannot be neglected. Another simulation performed in the present article has shown that the use of the latent heat of vaporization of free water (instead latent heat of vaporization of water in the product) is not significant to describe drying of bananas.

Many times the product ρc_p in the transient term is considered constant in the equation of heat conduction (Mariani et al., 2008; Ramsaroop and Persad, 2012; Lemus-Mondaca et al., 2013). This consideration made it possible to determine an analytical solution for the equation of heat conduction in a hemispherical shell geometry (Ramsaroop and Persad, 2012). However, the results obtained in the present paper indicated that such consideration worsened the results of model proposed and, consequently, should be avoided.

4. Conclusion

Liquid diffusion model, with vaporization of water at surface of the bananas, was assumed in this article. Model included mass and thermal diffusivities as variable properties, boundary conditions of the third kind, and also included the shrinkage. Using an infinite cylinder to represent the bananas, this model well described the mass and heat transfer during the drying process.

In the description of mass migration, the determination coefficients were greater than 0.99960 for all drying air temperatures. Also, it can be concluded that the expressions obtained for $D(M)$, as well as the values for convective mass transfer coefficients are adequate to describe mass migration during drying of bananas. On the other hand, in the description of heat transport, the determination coefficients were greater than 0.98000. In addition, the values obtained for thermal diffusivity and convective heat transfer coefficient are compatible with the expected values for mature bananas submitted to the experiments performed. Thus, it can be concluded that the expressions obtained for $\alpha(\overline{M})$ as well as the values for convective heat transfer coefficients are adequate to describe heat transport during drying.

It was possible to realize that the exclusion of the energy used to heat the vapor from the surface temperature up to the drying air temperature does not influence the description of the heating of bananas in a significant way. However, the exclusion of latent heat of vaporization strongly worsened the obtained results. On the other hand, the use of the value of the latent heat of vaporization of free water in the simulations (instead latent heat of vaporization of water in the product) is not significant to describe the process. Finally, it was concluded that the consideration of the product ρc_p with a constant value in the equation of heat conduction should be avoided.

Acknowledgment

The first author would like to thank CNPq (Conselho Nacional de Desenvolvimento Científico e Tecnológico) for the support given to this work and for his research grant (Process Number 301697/2012-4).

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